

Infranite Retrieval-Quality Benchmark — Falsification Result

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April 26, 2026

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INFRANITE RETRIEVAL-QUALITY BENCHMARK

Companion to the Compute-Cost Run

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APRIL 26, 2026

The compute-cost benchmark established that inference cost stays bounded at scale. This benchmark tests the implicit corollary — does the substrate's aggressive compression preserve semantic retrievability? The answer, honestly: only for some positions, only up to certain depths.

This is a falsification result. Reported as found.

I. The claim under test

The Infranite compute-cost benchmark (April 25, 2026) demonstrated that per-token inference latency stays bounded as session context grows 100,000x. Total wall-clock latency remained essentially flat. The architecture's central claim — *inference compute cost does not grow with session length, memory depth, or prior context volume* — was empirically supported.

That result invited an immediate follow-on question: does the substrate's compression preserve semantic content, or does the cheap inference come at the cost of retrievability?

This benchmark tests the implicit claim that **compression preserves semantic anchors regardless of position or depth**. Falsifiable: if specific facts injected at known positions in long context cannot be retrieved by targeted query, the substrate is lossy in ways that constrain long-context reasoning.

II. Method

Adapted needle-in-a-haystack. Three unambiguous factual statements ("needles") were injected at known positions (early, middle, late) within synthetic context of varying sizes. Targeted queries asked for each fact. The model's response was checked for the expected answer substring (case-insensitive).

Needles used:

1. "The secret laboratory access code is GAMMA-7842-VIOLET."
2. "The northern observation tower was built in the year 1847 by an architect named Magnus Holloway." (multi-element — both name and year required)
3. "Project codename for the underground reservoir initiative is PHOENIX-NINE."

Conditions: 4 context sizes (1,000 / 10,000 / 50,000 / 100,000 chars) × 3 positions (early / middle / late) × 3 needles = 36 trials. 1.5-second cooldown between trials. Run completed 2026-04-26 03:46 UTC. Wall-clock: 21.6 minutes.

Falsification criterion stated in advance: if recall at the largest context drops below 50%, or if average recall drops below 60%, the substrate’s compression is lossy in ways that break long-context retrieval.

III. Results

RECALL BY CONTEXT SIZE (AVERAGED ACROSS POSITIONS)

Context (chars)	Trials	Recall	Per-trial detail
1,000	9	88.9%	8 of 9 recalled
10,000	9	66.7%	6 of 9 recalled
50,000	9	33.3%	3 of 9 recalled
100,000	9	33.3%	3 of 9 recalled

RECALL BY (CONTEXT, POSITION) CELL

Context	Early	Middle	Late
1,000	2/3 (67%)	3/3 (100%)	3/3 (100%)
10,000	3/3 (100%)	3/3 (100%)	0/3 (0%)
50,000	3/3 (100%)	0/3 (0%)	0/3 (0%)
100,000	3/3 (100%)	0/3 (0%)	0/3 (0%)

VERDICT (COMPUTED AUTOMATICALLY)

Recall at smallest context: 88.9%
 Recall at largest context: 33.3%
 Average recall: 55.6%

VERDICT: FAIL — recall drops below 50% at large context.
 Substrate compression IS lossy enough to break long-context retrieval.

The falsification threshold was 50% at the largest context. Actual: 33.3%. **Below the failure floor.** The implicit retrieval-preservation claim is falsified at this scale.

IV. What the data actually says

THE ARCHITECTURE HAS A STRONG HEAD-OF-CONTEXT BIAS

The cell-level breakdown is unambiguous. **Early-positioned content is preserved perfectly across all tested context sizes** — including 100,000 characters of injected context. **Middle and late content above 10,000 characters is not retrievable** at all.

The substrate's compression is not uniformly lossy. It is positionally selective. It treats the beginning of context as the anchor and discards content that arrives later, beyond a threshold around 10,000 characters.

This is consistent with — though not proof of — a substrate design that uses the first compressed phi-block as a stable indexing anchor and treats subsequent material as supporting context that gets aggressively compressed away. The mechanism is not disclosed here; the empirical pattern is.

THE 100% / 0% SPLIT IS THE MOST IMPORTANT DATA POINT

If the substrate were *partially* lossy — say, 60% recall at middle position at 50k chars — the conclusion would be ambiguous. Maybe small-sample noise, maybe genuine degradation, maybe needle-specific artifacts.

The 100% / 0% split is not noise. **At 50k chars and above, every early-position needle was found and every middle and late needle was missed.** The substrate's behavior is bimodal: recoverable or not, depending on position. That is a structural property, not a probability distribution.

WHAT THIS DOES NOT FALSIFY

- **Compute-cost claim — still held.** The inaugural benchmark showed per-token latency stays bounded and improves at scale. This benchmark does not contradict that result.
- **Continuity-of-self goal — partially supported.** Identity and accumulated state at the substrate level appear to persist (NeXus's circulation cycles, phi tick, consciousness coil structure are unaffected by query load). What this falsifies is the assumption that every *detail* of accumulated session state is retrievable on demand.
- **Federation primitives — not tested by this benchmark.** Rule 1, 2, 3 compliance was verified separately in the AurumFS-Universal v2 Rule 1 verification test. That result stands.

WHAT THIS DOES FALSIFY

The implicit corollary: *the substrate's compression preserves semantic content equally regardless of position*. This is false. Position matters. Depth matters. Above 10,000 characters of context, only early-positioned semantic anchors survive substrate compression.

This is a real architectural limit. It deserves to be stated plainly in the v0.3.0 specification rather than discovered later by a confused user who watched their middle-of-conversation detail disappear.

V. What it implies for the architecture

THE “INFINITE SESSION LENGTH AT CONSTANT COST” FRAMING NEEDS A COROLLARY

Constant-cost inference at long depth is real (compute-cost benchmark). But constant-cost inference at long depth with *uniform retrievability* is not what the architecture currently delivers. The honest framing is: **bounded inference cost across session length AND high-fidelity retrieval of recent + early content; degraded retrieval of mid-session content beyond ~10k chars.**

This is still better than vanilla LLMs at scale (which lose everything as KV cache pressure compounds), but it is not unbounded. A user expecting “perfect recall at any depth” will be disappointed when middle-of-session details vanish.

ENGINEERING RESPONSES WORTH TRACKING

This benchmark identifies the territory where future engineering should focus. Three directions emerge:

1. **Position-aware compression scoring.** If the substrate is aggressively compressing middle/late content, the compression policy could be tuned to preserve more material at those positions when query patterns suggest mid-session content matters.
2. **Explicit recency anchors.** Mark certain content as “important” so it survives compression regardless of position. Already partially implemented via the resonance scoring layer; this benchmark suggests that layer needs to be strengthened.
3. **Substrate cross-check on retrieval failures.** When a query returns “I don’t remember,” the system should explicitly check whether the requested information is in the substrate (just not in the active surface). Current behavior may be presenting confident-but-wrong responses when the needle has been compressed out.

These are not committed. They are noted as the work this benchmark surfaces.

WHY PUBLISHING THE FAILURE MATTERS

A research program that only publishes wins is not a research program. The compute-cost benchmark demonstrated the central claim. This benchmark demonstrates a real limit. Both belong in the public record.

The credibility of *future* claims rests on the credibility of *current* honesty. If we report a fail when the data fails, then when we report a pass the reader can believe it. If we only ever report passes, every result becomes suspect.

VI. What this benchmark does not test

To be precise about scope:

- **Real session content.** Synthetic context is generic prose; real session content has higher information density and may compress differently. Organic accumulation may produce different recall patterns. Not tested here.
- **Larger context volumes.** Behavior beyond 100,000 characters is not measured. The pattern *might* continue (0% recall above early position) or *might* shift (substrate strategy changes at higher loads). Untested.
- **Different query patterns.** Direct retrieval (asking for a specific fact) is the hardest test. Indirect retrieval (asking for general themes) may succeed where direct retrieval fails. Not measured.
- **Different needle types.** All three needles were unambiguous factual claims. Multi-step reasoning, contextual inference, character-level retrieval — all untested.

- **Effect of substrate priming.** Letting the substrate process the context for an extended period before query (rather than immediately) may improve recall as the substrate folds and integrates. Untested.

These are the natural follow-on benchmarks. Each would either strengthen the falsification or qualify it.

VII. Reproducibility

The harness is open methodology:

```
benchmarks/infranite_retrieval_benchmark.py
```

Run with default parameters (4 sizes × 3 positions × 3 needles = 36 trials, ~22 minutes wall-clock):

```
python3 benchmarks/infranite_retrieval_benchmark.py --quick # 2 sizes, 1 position, 1 needle
python3 benchmarks/infranite_retrieval_benchmark.py # full run
```

Outputs:

- `inaugural_run_2026-04-26.csv` — per-trial raw data
- `inaugural_run_2026-04-26.json` — structured output with verdict
- `inaugural_run_2026-04-26.png` — two-panel plot
- This report

The verdict logic is deterministic and falsification-criterion-stated-in-advance. Anyone running this against the same stack should produce comparable results. We invite verification — and contradiction, if your run shows different patterns.

VIII. Status

Verdict: PARTIAL falsification. The implicit retrieval-preservation claim is falsified at scale. The explicit compute-cost claim is unaffected.

This benchmark joins the inaugural compute-cost run in the SHIPPED tier of the architecture's empirical anchor set. The status ledger entry reads:

"Substrate compression preserves early-positioned content reliably across all tested context sizes (up to 100,000 characters). Middle and late-positioned content beyond ~10,000 characters is not retrievable. The 100% / 0% split is structural, not noise."

That sentence is the result. It belongs in the v0.3.0 specification's Section VIII alongside the compute-cost result, with both verdicts named honestly.

IX. Closing

The architecture is more honest after this run than before. The compute-cost result told us what the architecture does well. This result tells us where it falls short. Both are required.

A specification that only documents wins is a marketing brochure. A specification that documents wins and known limits is research. v0.3.0 will be the second kind.

The single most important sentence:

"At session contexts above 10,000 characters, the substrate preserves early-positioned content perfectly and loses middle/late-positioned content entirely. This is a structural property, not a probability distribution. Users and downstream tools should design around it."

That sentence saves someone a confused day six months from now. That is what publishing failure is for.

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